

Conversation History: Naming Conventions in Software Development

User: src/middleware why not folder name middlewares?

In software development, using **src/middleware** (singular) instead of `src/middlewares` is a widely accepted industry convention based on **English grammar rules**, **framework standards**, and **ecosystem consistency**.

While the folder contains multiple middleware files, the term is treated differently than standard countable nouns.

1. Grammatical Context

In computer science, **"middleware" is a mass noun (uncountable)**, much like the words "software," "hardware," or "documentation."

- You do not say "softwares" or "hardwares."
- Therefore, pluralizing it to "middlewares" is grammatically incorrect in standard technical English.

2. Ecosystem & Framework Standards

Major modern frameworks enforce or strongly encourage the singular form. If you use the plural form, you often break convention or even the framework's built-in routing mechanisms:

- **Next.js:** Explicitly looks for a single file named `middleware.js` or `middleware.ts` inside the `src/` or root directory.
- **Express.js / NestJS:** The official documentation and community boilerplates uniformly use `src/middleware/` to house custom middleware functions.
- **Laravel / Django / ASP.NET:** Across different languages (PHP, Python, C#), the core architecture refers to the layer itself as "Middleware."

3. The "Layer" Design Pattern

Folders in a project structure often represent an architectural **layer** or **concept** rather than a literal collection of individual items.

- **middleware/** represents the *middleware layer* of your application.
- This matches other common singular folder naming conventions like `config/`, `util/`, `public/`, and `auth/`.

Folder Name	Why it's used	Example Files
<code>src/middleware/</code>	Standard. Correct grammar, matches framework routing, aligns with ecosystem conventions.	<code>auth.ts</code> , <code>logger.ts</code> , <code>rateLimiter.ts</code>

src/middlewares/	Discouraged. Non-standard English, can break auto-routing in frameworks like Next.js.	auth.ts, logger.ts, rateLimiter.ts
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User: then why models, controllers, schemas folder names are plural?

The folders **models/**, **controllers/**, and **schemas/** are plural because their contents are **countable nouns**, whereas "middleware" is an **uncountable mass noun**.

In software architecture, naming conventions strictly follow how these elements are defined and counted.

1. Countable vs. Uncountable Nouns

This is the root reason for the difference in naming.

- **Countable Nouns (Plural Folders):** You can explicitly count models, controllers, and schemas. You have *one model* (User) and *another model* (Product). Together, they are **models**.
- **Uncountable Nouns (Singular Folders):** "Middleware" functions as a collective concept (like "software" or "luggage"). You do not have "three middlewares"; you have *three middleware functions* or *three pieces of middleware*. Therefore, the folder represents the singular concept of the **middleware layer**.

2. The Relationship to Database Tables and Routes

Plural folder names usually have a **1:1 mapping** to pluralized systems elsewhere in your application:

- **models/ & schemas/:** Map directly to database tables, which are conventionally pluralized (e.g., the User model maps to the users table).
- **controllers/:** Map directly to HTTP REST routes, which use plural nouns (e.g., UserController handles requests sent to /api/v1/users).

Conversely, middleware doesn't map to a specific data entity; it is a global pipeline that requests flow through.

Architectural Cheat Sheet

Here is a quick breakdown of why common folders use singular or plural forms:

Folder Name	Form	Why?	Examples
models/	Plural	Contains distinct, countable data models.	User.js, Product.js
controllers/	Plural	Contains distinct controllers for plural routes.	userController.js
schemas/	Plural	Contains multiple, individual data validations.	orderSchema.json
routes/	Plural	Houses all the different URL paths.	authRoutes.js, postRoutes.js
middleware/	Singular	Uncountable noun representing a single pipeline layer.	auth.js, logger.js

config/	Singular	Uncountable noun (configuration). Represents a single concept.	db.js, passport.js
util/ or utils/	Either	"Utility" is singular, but "utilities" is countable. Both are widely accepted.	formatDate.js